COLLEGE ONLINE VOTING SYSTEM

A Project Report | Bachelor of Computer Science

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ABSTRACT

13	10	76.9%	22
Registered Voters	Votes Cast	Voter Turnout	Test Cases Passed

This a secure, web-based college election platform built with Python Flask, MySQL, HTML5, CSS3, JavaScript, and Chart.js. It replaces paper-based elections with a fully digital system accessible from any internet-enabled device.

Key Features:

- Auto-generated unique Voter IDs in VTR-XXXXXXXX format
- Real-time Chart.js bar charts on admin dashboard
- One-vote-per-student enforcement at application AND database levels
- SHA-256 password hashing & session-based access control
- Candidate TOGGLE functionality (active/inactive)
- Professional dark-navy CIVITAS branding

EXISTING SYSTEM & PROBLEMS

Physical Attendance Required

Students must visit polling booths — those absent on election day are completely excluded from voting.

Time & Resource Intensive

Requires printing ballots, staffing booths, and hours of manual counting by committee members.

Error-Prone Manual Counting

Miscounts and spoiled ballots affect accuracy; no double-check mechanism available.

No Real-Time Monitoring

Results only available after complete counting — zero transparency during the election process.

Security Vulnerability

Physical ballots vulnerable to tampering, damage, or destruction before counting is complete.

Environmental Impact

Large-scale paper printing and disposal contributes to unnecessary environmental degradation.

PROPOSED SYSTEM

Web-Based Digital Platform

Accessible from any internet-enabled device — no physical presence required for casting votes.

Register Number Authentication

Students log in using college register number with SHA-256 password hashing for security.

Auto-Generated Voter IDs

Unique VTR-XXXXXXXX format Voter ID automatically issued upon successful student registration.

One-Vote Enforcement

UNIQUE database constraint + application-level logic ensures each student can vote only once.

Real-Time Admin Dashboard

Live vote distribution bar charts via Chart.js; admin sees standings as votes are cast.

Candidate Management

Admin can add, edit, delete, and toggle candidates active/inactive during the election.

SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS	
OS	Windows 10 / 11
Backend	Python 3.8+ & Flask 2.0+
Database	MySQL 8.0
Frontend	HTML5, CSS3, JavaScript
Browser	Chrome / Firefox
IDE	VS Code / PyCharm

HARDWARE REQUIREMENTS	
Processor	Intel Core i3 (2.0 GHz+)
RAM	Minimum 4 GB
Storage	500 MB free space
Display	1280×720 resolution
Network	1 Mbps+ internet
Input	Keyboard & Mouse

SYSTEM MODULES

01

Student Registration

Register with details; SHA-256 hashed password; auto-generate unique VTR-XXXXXXXX Voter ID.

02

Student Login

Authenticate with register number; create session; redirect to personalized dashboard.

03

Student Dashboard

View voter profile, election schedule, active candidates, and voting status dynamically.

04

Voting Module

Cast one vote securely; atomic transaction; UNIQUE DB constraint prevents double-voting.

05

Admin Authentication

Admin login with predefined credentials; create admin session for full system access.

06

Admin Dashboard

Real-time stats: total voters, votes cast, active candidates + Chart.js bar chart visualization.

07

Candidate Management

Add, edit, delete candidates; toggle active/inactive status; upload candidate photo.

08

Election Management

Configure election title, description, start/end dates; auto-control voting availability.

09

Database Module

MySQL ACID compliance; UNIQUE constraints; Foreign keys; secure data storage & retrieval.

10

Visualization Module

Chart.js responsive bar charts; vote data passed as JSON from Flask backend.

TECHNOLOGY STACK

Python	**Flask**	**MySQL**	**HTML5 / CSS3**
Backend Language	Web Framework	Database	Frontend Markup
JavaScript	**Chart.js**	**SHA-256**	**Jinja2**
Client Logic	Visualizations	Password Hashing	Template Engine

SYSTEM DESIGN — MVC ARCHITECTURE

MODEL

MySQL Database Tables + Python Data Access
students | candidates | votes | admin

VIEW

12 Jinja2 HTML Templates
Dark-Navy CSS Theme

CONTROLLER

Flask Route Handlers in app.py
@app.route() decorators — HTTP request processing

Student Module

Admin Module

FUTURE ENHANCEMENT

Blockchain Integration

Implement blockchain for immutable, cryptographically verifiable vote records.

Mobile Application

Develop native iOS and Android apps for easier mobile access and notifications.

OTP-Based Authentication

Add two-factor authentication with OTP via email/SMS for stronger security.

Multi-Election Support

Allow simultaneous management of multiple elections in a single platform.

Analytics & Reporting

Generate detailed PDF/Excel reports of election results and voter statistics.

Biometric Verification

Integrate face recognition or fingerprint scan for voter identity verification.

CONCLUSION

This is successfully delivers a modern, secure, and accessible digital election platform for college institutions. By replacing inefficient paper-based voting with a Flask + MySQL web application, it ensures real-time transparency, strong security, and high voter participation — validated with 76.9% turnout and all test cases passing.

The system is production-ready and deployable for any college election scenario.

- Secure SHA-256 authentication
- One-vote-per-student enforced
- Real-time Chart.js visualization
- MVC clean architecture

THANK YOU

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